Formalizing AI Scientist Systems: A Component Framework and Theoretical Analysis

Jaesol Shin
WeDataLab
ysys143@gmail.com
https://github.com/ysys143

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Abstract

The field of *AI Scientist* systems—agents that generate hypotheses, execute experiments, and synthesize knowledge—lacks a shared formal vocabulary. We propose Servo (S, G, E, V, M, π), a six-component taxonomy comprising the **S**earch space, hypothesis **G**enerator, **E**xperiment executor, **V**alidator, **M**emory, and search **P**olicy (π), grounded in Partially Observable Markov Decision Process (POMDP) and Bayesian Experimental Design (BED) theory. The framework's core contribution is three diagnostic separations that current systems conflate: the observation likelihood from the validator as a reward channel; calibrated gating (V_{gating}) from mere layer presence (V_{present}); and episodic from semantic memory. Appendix A uses standard Bayesian results to delimit these distinctions in fixed, idealized settings: posterior consistency requires conditions on the observation model and data-generating process; a validator coupled into a normalized likelihood can change its pseudo-true target; and a novelty component absent from every available channel is non-identifiable. These bounded analytical observations clarify the vocabulary but neither constitute new general theorems for adaptive AI Scientist systems nor certify empirical prevalence. We apply the framework to six core systems and seven domains and identify three open problems with falsification criteria: the lack of validated automated novelty gates, the BED–practice gap for search policies, and the experiment fidelity gap (no principled criterion for escalating from computational proxy to physical validation). The cross-system survey is offered as provisional structured annotation; the proposed association between gating calibration and trustworthy closure remains untested because the six-system outcome coding is partly entangled with V; it is a hypothesis for future independent evaluation, not an empirical finding.

1 Introduction

AI Scientist systems—agents that independently generate hypotheses, execute experiments, and synthesize knowledge—have proliferated rapidly [Boiko et al., 2023, Lu et al., 2024, 2026, Feng et al., 2026, Liu et al., 2026], yet the field lacks a shared formal vocabulary. Systems range from tool-augmented synthesis agents (Coscientist [Boiko et al., 2023]) through closed-loop manuscript-writing pipelines (The AI Scientist [Lu et al., 2024, 2026]) to natural-language mathematics agents

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(Aletheia [Feng et al., 2026]), each introducing bespoke terminology that makes systematic cross-system comparison difficult. Without a shared decomposition, questions that bear directly on deployment remain unanswered: *What determines whether a system can operate in a closed loop? Why does autonomous novelty evaluation fail consistently? What is an appropriate information-seeking search policy?*

We propose Servo[1] (S, G, E, V, M, π), a formal framework grounding AI Scientist systems in POMDP and Bayesian Experimental Design theory (Section 4). We apply it to analyze six core systems, an artifact infrastructure proposal, and instantiations across seven scientific fields (Sections 5, 6); and identify three open problems with falsification criteria (Section 7). The analysis reveals a recurring diagnostic distinction: systems that mechanically close the loop differ sharply in whether their acceptance gate, novelty judgment, and memory structure can support trustworthy discovery claims; we develop the supporting theory in Appendix A. The contribution is thus a diagnostic vocabulary that keeps mechanical closure, validator calibration, novelty judgment, and memory separable rather than collapsing them into a single notion of autonomy.

Scope. We include systems whose primary goal is automated scientific *discovery* through hypothesis-experiment-validation cycles, and exclude specialized predictors without a hypothesis-validation loop (e.g., AlphaFold [Jumper et al., 2021]). Within that scope we analyze widely-cited representative systems rather than an exhaustive enumeration; this convenience sampling introduces selection bias that we flag in Section 8.

2 Background

Following Kaelbling et al. [1998], we model AI Scientist systems as *partially observable Markov decision processes*: the true state s (unknown structure of nature) is inaccessible; the system maintains a belief $b(s)$ updated by experimental observations. Formally, an AI Scientist POMDP is the tuple $(\mathcal{S}, \mathcal{A}, T, R, \Omega, \mathcal{O}, b_0, \gamma)$ with action space $\mathcal{A}$ (hypothesis generation and experiment execution) and reward R (scientific value of discovery); the belief over s is updated by Bayes' rule, and π acts on that belief rather than the inaccessible state s. Three properties distinguish this from standard control: R cannot be pre-specified (significance is not defined a priori); $\mathcal{S}$ is unbounded; and T is non-stationary (accumulated knowledge changes the exploration frontier). We therefore use the POMDP as an organizing formalism, not a fully specified model of open-ended science. Appendix A instead separates observation, reward, and novelty channels in fixed idealized settings; it does not establish posterior consistency for the adaptive, open-ended systems surveyed here. Throughout, three distinct senses of "loop closure" should be kept separate: (i) posterior concentration in a specified statistical model; (ii) *computational* closure, where feedback from V flows back into G; and (iii) *trustworthy* closure, where the acceptance gate reliably tracks truth. The appendix clarifies conditions relevant to (i); the survey concerns (iii); systems may achieve (ii) without (i) or (iii). Bayesian Experimental Design [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] provides a normative criterion for the information objective of the search policy π: select d^* maximizing Expected Information Gain (EIG),

$$d^* = \arg\max_{d \in \mathcal{D}} \; \mathrm{EIG}(d) = \arg\max_{d} \; \mathbb{E}\left[H(\theta) - H(\theta \mid y, d)\right] \tag{1}$$

where $H(\theta)$ is prior entropy over parameters θ and $H(\theta \mid y, d)$ is expected posterior entropy after observing y under d. EIG can be approximated via variational methods [Foster et al., 2019], amortized policies [Foster et al., 2021], and RL [Blau et al., 2022]. Domain-specific systems already use active-learning approximations—the Robot Scientist's experiment selection [King et al., 2004] and GNoME's DFT-in-the-loop active learning [Merchant et al., 2023]—but recent open-ended LLM-based systems maintain no explicit posterior or stated EIG objective; that narrower gap is the one we identify in Section 7.

[1] We name the framework after the *servomechanism*: as a servo loop converges on a target by acting on a feedback (error) signal, an AI Scientist closes the discovery loop only when its validator V supplies a reliable feedback signal—the framework's central concern.

3 Related Work

AI Scientist systems. Prior work [Lu et al., 2024, 2026, Schmidgall et al., 2025, Boiko et al., 2023] provides qualitative characterizations of individual systems but no shared framework for cross-system comparison; idea-generation studies [Si et al., 2024, Baek et al., 2024] establish quality baselines without unifying the full discovery loop; knowledge-graph-driven ideation systems [Gu and Krenn, 2024, Lee et al., 2025] further show that structuring semantic memory as a relational corpus can help surface and rank higher-interest research ideas, but equally without closing the loop.

Autonomous-science frameworks. Closed-loop autonomous discovery predates the LLM era: the Robot Scientist [Sparkes et al., 2010] integrated hypothesis generation, robotic experimentation, and statistical validation in yeast functional genomics, instantiating all six Servo components—including an active-learning policy [King et al., 2004]—in physical wet-lab form more than a decade before LLM-based systems. More recent unified frameworks include NovelSeek [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025], a closed-loop multi-agent system spanning idea generation to verification across twelve scientific tasks, and Co-Scientist [Gottweis et al., 2026], a multi-agent hypothesis-generation system with experimental biomedical validation. We therefore position Servo not as the first unified framework but as a formal vocabulary making validator completeness, the computational-to-physical boundary, and the POMDP/BED relationship explicit for the LLM-agent subfield. Its diagnostic value is illustrated, not independently tested, at two points. First, standard POMDP classification and NovelSeek's multi-agent framework distinguish loop-closed from loop-open but do not separate V_{present} (a calibrated layer exists) from V_{gating} (the decisive gate is calibrated); both code AI Scientist (Nature 2026) as closed-loop, but Servo's V-layer decomposition locates the failure—the gating layer is biased, not absent. Second, existing POMDP treatments collapse loop closure into a binary, while Servo's three-way separation (posterior concentration, computational closure, trustworthy closure) identifies systems such as Agent Laboratory that close a computational loop without trustworthy gating.

Decision-theoretic foundations. The POMDP framework [Kaelbling et al., 1998] and BED [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] are classical, with a long history of Bayesian-optimal experiment selection in automated science—most directly the Robot Scientist's active-learning policy [King et al., 2004]; we connect POMDP (structure) to BED (normative policy) for the LLM-agent setting rather than claiming priority for the idea. The reward-specification and specification-gaming literatures study unreliable reward signals; Servo extends this lens by stratifying V into reliability levels tied to specific closed-loop failure modes.

Formal models and artifact infrastructure. Taniguchi et al. [2025] propose Collective Predictive Coding (CPC-MS) as a social model of science, recasting scientific activity as decentralized Bayesian inference across a community. CPC-MS and Servo are complementary: CPC-MS provides a *collective* and *social* model in which peer review is modeled as a Metropolis-Hastings acceptance (consensus) process; Servo provides the *single-system* and *modular* decomposition that must be instantiated at each node. The three open problems we identify constitute core implementation challenges for such a collective, viewed through Servo. Liu et al. [2026] address the artifact layer — output format and verification protocol — rather than the discovery loop itself; we position ARA as a V infrastructure contribution within Servo (the Seal automates V_{syntax}, V_{semantic}, and $V_{\text{empirical}}$), discussed in Sections 5.1 and 7.

4 The Servo Framework

We define an AI Scientist system as a tuple (S, G, E, V, M, π) with the following components:

$$\text{Servo} = (S,\ G,\ E,\ V,\ M,\ \pi) \tag{2}$$

S**: Search Space** The set of candidate hypotheses, programs, molecules, experimental conditions, or other objects subject to exploration. S may be static or dynamically expanding (open-ended).

$G : M \times S \to h$**: Hypothesis Generator** A function that proposes new candidate $h \in S$ conditioned on accumulated memory M and the current state of the search space. G may involve LLM generation, evolutionary mutation, or symbolic search.

$E : h \times P \to o$**: Experiment Executor** A function that executes hypothesis h under protocol P to produce observation o. The fidelity of E ranges from fast computational simulation (low cost, potential surrogate error) to physical wet-lab execution (high cost, ground truth).

$V : o \to r$**: Validator** A function that assigns a reward signal r to observation o. We distinguish four layers of V:

$$\begin{aligned}
V_{\text{syntax}} &: o \to \{\text{pass, fail}\} && \text{(deterministic, fast)} \\
V_{\text{semantic}} &: o \to r \in [0,1] && \text{(LLM-based, stochastic)} \\
V_{\text{empirical}} &: o \to r && \text{(reproduction-based)} \\
V_{\text{human}} &: o \to \{\text{novel, significant}\} && \text{(human-delegated)}
\end{aligned}$$

M**: Memory** Structured as (M_e, M_s, M_p). M_e (episodic) stores individual records (h_i, o_i, r_i); M_s (semantic) stores distilled knowledge patterns; M_p (procedural) stores learned protocols. The structure of M directly determines the quality of G and π.

$\pi : b \times M \to h$**: Search Policy** The decision rule for selecting the next hypothesis $h \in S$. The belief state b is updated after each experiment. When the next decision encodes a design d, a myopic BED rule may be written $\pi_{\text{EIG}}(b, M) \in \arg\max_{d \in \mathcal{D}} \text{EIG}(d)$ (Equation 1); this is a one-step action rule, not a globally optimal sequential policy. In practice, current systems use policies ranging from reactive (ReAct) to greedy to tree search.

The operational loop is:

$$\begin{aligned}
&\pi(b, M) \;\to\; G(M_s, S) \;\to\; E(h, P) \;\to\; V(o) \\
&\qquad\qquad \to\; M \mathrel{+}= (h, o, r) \;\to\; b \text{ update} \;\to\; \text{repeat}
\end{aligned} \tag{3}$$

We further introduce $H \in [0, 1]$ (distinct from the entropy function $H(\theta)$ in Eq. 1) as an auxiliary parameter denoting the degree of human intervention: $H = 0$ for fully autonomous systems and $H = 1$ for systems in which humans assume the role of G or π.

V-completeness as a design dimension. The layers form a partial order under reliability: V_{syntax} deterministic; V_{semantic} stochastic and bias-prone; $V_{\text{empirical}}$ from well-calibrated (DFT) to poorly calibrated (GNN surrogates); V_{human} is the novelty channel used by the surveyed systems but is rate-limiting; and V_{formal} (proof kernels, type-checkers) is the mechanically-decidable ideal limit, used for mathematics (formal) in Table 2. We define V *completeness* as the highest layer both accessible and reliably calibrated; accessibility without calibration does not count (Agent Laboratory's biased reviewer yields only imperfect completeness). We treat calibration of the gating layer as an operative design variable—the holistic ordinal being exploratory (Appendix B)—because it can limit what G is rewarded for and what progress the acceptance channel reports to π. This does not prevent learning through a separately informative observation likelihood. Throughout, *calibration* denotes the weaker property that the gating layer is not systematically biased, in the operational sense of the released coding protocol; no surveyed system reports probabilistic calibration (reliability curves, expected calibration error, Brier score), so that stronger property is neither claimed nor coded here. What matters for acceptance feedback is not that a calibrated layer is *present* but that the layer which *gates acceptance* is calibrated: a calibrated check can be neutralized when a biased semantic or social layer decides acceptance (§8). In this sample, the cross-domain pattern is associated with gating-validator calibration alongside differences in G, E, M, and π; Appendix A gives bounded analytical observations that keep the relevant channels distinct.

The validator as a reward channel. Treating V as distinct from the observation likelihood, Appendix A records three analytical separations. Posterior consistency belongs to the observation model and data-generating regime; a validator changes the inferential target only when it is coupled into a properly normalized update likelihood (or indirectly changes future designs); and a novelty component invisible to every available channel is non-identifiable. These observations diagnose where validator design can matter. They do not prove that a complete validator is universally necessary or sufficient for trustworthy autonomous discovery; that remains the paper's provisional design hypothesis.

5 Analysis of Core AI Scientist Systems

We apply Servo to the core systems in Table 1 and one artifact infrastructure proposal, synthesizing the key dimensions; in this small sample, the systems that reach closed-loop operation are also those with more complete validators, although they also advance G, E, and π (we revisit this interaction in Section 8).

	Robot Sci.	**Coscientist**	**AI Sci. (2024)**	**AI Sci. (2026)**	**AgentLab**	**NovelSeek**
Novelty assessment	Predefined	Human	Biased auto	Biased score; non-gating	Biased auto	None
Loop closed?	Yes (wet-lab)	Partial	Yes (greedy)	Yes (tree)	Partial	Yes (comp.)
π type	Active learn.	ReAct	Greedy	Tree search	Human/seq.	Feedback
V layers	V_e+V_h	V_y+V_e+V_h	V_y+V_s^*	V_y+V_s^*+V_e	V_y+V_s^*+V_h	V_y+V_e
M structure	$M_e + M_s$	Ephemeral	M_e	$M_e + M_s$	M_e	M_e
H (approx.)	0.25	0.75 (G)	0.1	0.1	0.8 (G+π)	0.25

Table 1: Comparative Servo analysis of core AI Scientist systems, including the pre-LLM Robot Scientist [Sparkes et al., 2010] and the recent unified framework NovelSeek [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025], which later versions of that preprint retitle *InternAgent*. V_y=syntax, V_s=semantic, V_e=empirical, V_h=human; an asterisk marks a layer the source itself reports as systematically biased. The novelty-assessment row describes how novelty is scored or supplied; it does not assert that the scorer advances search. It corresponds to the legacy `novelty_gate` field in `analysis/systems.csv`, whose name should not be confused with operational `V_gating`. The Robot Scientist's active-learning policy is documented in [King et al., 2004]; the 2010 review [Sparkes et al., 2010] describes the system without stating a selection objective. For AI Scientist (Nature 2026), the biased automated reviewer is an assessment instrument but not an operational gate, so `V_gating=0` in the released coding; the external workshop review its outputs received sits outside the loop and is not counted as a validator layer.

Across these four LLM-based end-to-end systems (Coscientist, the two AI Scientist systems, and Agent Laboratory), those that close the loop are also those that upgrade or add validators; G or π advances are not absent—several also upgrade ideation, tool use, and search—so validator reliability is one recurring diagnostic bottleneck associated with whether such advances are exploited in a closed loop, not an isolated causal constraint. Coscientist has only constrained task-level feedback (measured outcomes inform later choices [Boiko et al., 2023]); its loop is *partial*, with novelty/significance validation human-mediated. The AI Scientist (Sakana) adds V_s via a 5-ensemble GPT-4o reviewer (balanced accuracy 0.65 predicting ICLR 2022 accept/reject labels, alongside a 0.66 human inter-reviewer consistency baseline measured on a different pool—the two are not the same quantity [Lu et al., 2024]), enabling the first *structurally* closed loop on a biased, uncalibrated V. The AI Scientist (Nature)[2] stacks V_y (execution checks), V_e (replication nodes with mean $\pm$ s.d.) and V_s (automated reviewer, balanced accuracy 0.69 [Lu et al., 2026]), giving the most automated layers in the class; its outputs were additionally sent to an external workshop review, which lies outside the loop and is not coded as a validator layer—though its automated reviewer is an evaluation instrument rather than a gate on the search (stage transitions are decided by a stage evaluator, a best-first policy and a compute budget, and the manuscripts submitted externally were selected by hand) and it exhibits a marked acceptance bias (false-positive rate 0.45 against 0.17 for human reviewers [Lu et al., 2026]) (§8). Agent Laboratory delegates V to V_s (NeurIPS-guideline LLM reviewer) but documents $+2.3$-point systematic over-estimation relative to the human PhD-student reviewers of the same AI-generated papers—adding a V layer does not guarantee completeness if the layer is unreliable [Schmidgall et al., 2025]. Four end-to-end systems, however, cannot establish general impossibility.

5.1 Artifact Infrastructure: Agent-Native Research Artifacts

Liu et al. [2026] address a complementary problem—what the *output* of the loop should be and how to verify it. Agent-Native Research Artifacts (ARA) replace the narrative paper with a four-layer ontology (`/logic`, `/src`, `/trace`, `/evidence`); the ARA Seal automates V_{syntax}, V_{semantic} (Rigor Auditor, 82.6% detection), and $V_{\text{empirical}}$ (sandboxed reproduction), reserving V_{human} for significance and novelty [Liu et al., 2026]. Motivating figures: only 45.4% of reproduction requirements are fully specified and 90.2% of extension cost is failed exploration [Liu et al., 2026]. The `/trace` layer externalizes M_e/M_s, but preserved traces both accelerated and anchored downstream agents—richer M_e does not monotonically improve π.

[2]The AI Scientist's (2024) and (2026) variants are its template-based and agentic-tree-search ("AI Scientist-v2" [Yamada et al., 2025]) modes, the latter its peer-reviewed Nature version [Lu et al., 2026].

6 Domain Analysis

Table 2 is an illustrative domain map: it uses the Servo vocabulary to expose how validation bottlenecks differ across seven scientific fields, with the observation–reward–novelty distinctions in Appendix A serving only as analytical diagnostics (per-domain analysis with sources in Appendix C). Systems analyzed span mathematics (formal: [Romera-Paredes et al., 2024, Xin et al., 2024]; NL: [Feng et al., 2026]), physics [Udrescu and Tegmark, 2020, Cranmer et al., 2020], chemistry [Bran et al., 2024, Sprueill et al., 2024, Kamber, 2026], materials [Merchant et al., 2023], biology [O'Donoghue et al., 2023], social science [Manning et al., 2024, Aher et al., 2023, Park et al., 2023a], and CS/algorithms [Real et al., 2020, Surina et al., 2025, Jaber and Jaber, 2026].

Domain	V type	V reliability	Loop	π	Primary bottleneck
Math (formal)	V_{formal}	Perfect oracle	Fully closed	RL/MCTS	Pretraining contamination
Math (NL)	$V_{\text{sem}} + V_h$	Imperfect	Closed (low quality)	Seq. refine	Hallucination; plagiarism
Physics	$V_{\text{emp}} + V_h$	Medium	None/partial	Deterministic	No cross-problem M
Chemistry	V_{emp} (surr.)	Medium	Comp. only	Beam	Wet-lab loop open
Materials	V_{emp} (DFT)	High (comp.)	Comp. closed	Uncert. samp.	DFT wall-time; synth. lag
Biology	$V_{\text{emp}} + V_h$	Low–med	Partial	None	V metrics insufficient
Social Sci.	Stat. test	Low–med	Partial	Factorial/ablation	Circular validity
CS / Algo.	V_{emp}	High	Fully closed	Evol./LLM	Interpretability absent

Table 2: Illustrative Servo domain mapping across seven fields (per-domain coding in `domain_systems.csv`). V reliability is an ordinal calibration label, not measured agreement with ground truth. Domain codings are single-coder interpretations of the cited primary sources; no inter-coder κ statistics are available for this table.

The pattern is largely shaped by V type. Formal mathematics has a binary V_{formal} that enables mechanically closed evaluation within the formalized task; natural-language proof reintroduces V_{human} (Aletheia: 6.5% audited-correct). Chemistry and biology close the computational loop but leave the physical loop open at the measurement boundary, where characterization stays human-delegated. Materials science is the surveyed domain whose policy most resembles active experimental design within a tractable computational surrogate: GNoME uses model uncertainty and DFT relaxations to close a computational loop for phase-stability prediction. But DFT approximates physical energies rather than being a fully calibrated oracle, and most predicted compounds still require synthesis and characterization [Cheetham and Seshadri, 2024]; the physical loop remains only partly closed. Social science suffers circular validity when G and E share a model, and CS/algorithms close the loop via reliable V_{emp} but lack interpretability.

7 Open Problems

Our analysis identifies three open problems that recur across the surveyed systems and illustrative domains. For each we state a *proposed* falsification test—a concrete design whose outcome would refute the claim—specifying what evidence would settle the question; none are executed here.

7.1 The Lack of Validated Automated Novelty Gates

No automated validator among the surveyed systems reliably evaluates novelty—every surveyed system, from Coscientist (no mechanism) to The AI Scientist Nature (three-layer validation), converges on this. Two failure modes are documented: LLM self-evaluation bias (systematic overestimation; hallucinated results) and pretraining contamination (systems may "discover" results implicit in training data, as Aletheia documents for Erdős problems). The ARA Seal's Rigor Auditor [Liu et al., 2026] achieves 82.6% average detection across five structural violation types but only 22% on orphaned-experiment violations; it cannot cross the novelty boundary. Concretely: Agent

Laboratory's LLM reviewer over-estimated paper quality by $+2.3$ points relative to human PhD students [Schmidgall et al., 2025]; The AI Scientist (Sakana) hallucinated ablation tables; and of 200 Erdős solutions audited (from 700 attempted), only 13 (6.5%) were rated meaningfully correct. A 45-expert evaluation likewise positions current AI reviewers as complements to, not substitutes for, human experts—strong on aggregate criticism yet sharing blind spots and overlapping far more than independent humans do [Kim et al., 2026]. This unreliability is not confined to novelty scoring: LLMs asked to reverse-engineer black-box systems—a canonical scientific inference task—fail to do so reliably even with systematic query access [Geng et al., 2025]. The Leiden Declaration [Alper et al., 2026], signed by over 3,000 mathematicians as of July 2026 (count from the signature site, `https://leidendeclaration.ai`; the declaration text itself states none), warns that AI-produced proofs may contain almost-invisible errors and provides a practical reason to retain expert review in current natural-language proof workflows. It does not establish that an informative novelty or significance channel must be human in principle.

Proposed falsification test: The thesis is refuted if an automated validator matches a panel of $\geq$3 domain experts on novelty discrimination—Cohen's $\kappa \geq 0.80$ against the expert majority—on a benchmark of published-versus-withheld findings whose training-set membership is fixed by a pre-training cutoff date, across $\geq$3 distinct domains. Constructing the contamination-controlled benchmark and the expert baseline is itself an open task; a recent benchmark takes a step by restricting to post-2024 findings to control contamination [Liu et al., 2025], though it evaluates inspiration-based decomposition rather than closing the expert-baseline gap.

Novelty, significance, and correctness are distinct properties that current systems keep apart when scoring but collapse at the decision point (The AI Scientist rates ideas on separate interestingness, feasibility and novelty scales, yet accepts or rejects a paper on a single aggregate `Overall` threshold [Lu et al., 2024]); a viable automated novelty V would need a contamination-controlled reference corpus, architectural separation between G and V, and a definition of "novel to the field" beyond lexical training-set membership (e.g. disruption or atypicality measures [Park et al., 2023b, Uzzi et al., 2013])—none of which current systems provide.

7.2 The BED–Practice Gap for π

Equation 1 specifies a one-step, myopic information criterion, yet no surveyed LLM-based system implements it: all deployed LLM-based policies are heuristic (ReAct, greedy, tree search, human-guided). The partial exception is GNoME [Merchant et al., 2023], whose deep-ensemble uncertainty estimates over a tractable GNN posterior can be read as a coarse EIG surrogate—but only because DFT provides a well-calibrated computational V and the hypothesis space admits a learnable probabilistic model, conditions absent from the LLM-based systems here, where the "posterior" is implicit in model weights [Foster et al., 2021]. BED-LLM [Choudhury et al., 2026] derives a usable EIG posterior from an LLM's predictive distributions, but for conversational queries rather than experiment design—so the gap is integration with reliable E and V, not impossibility. Concretely, every LLM-based system in Table 1 uses a heuristic π: Coscientist selects tools reactively, The AI Scientist (Sakana) filters ideas by archive similarity (which we code as greedy; the source states no selection objective), and Agent Laboratory follows a fixed schedule—none implements an explicit posterior-updating information objective. Recent falsification-driven approaches ground hypothesis selection in executable tests rather than free generation—AIGS selects hypotheses by their survival of automated refutation [Liu et al., 2024], and Popper runs sequential falsification experiments with statistical error control [Huang et al., 2025]—but neither computes or maximizes EIG: they substitute a falsification heuristic for the BED criterion rather than realizing it.

Proposed falsification test: The gap is refuted if a system explicitly computes and maximizes EIG over a posterior $p(\theta \mid y)$ whose hypothesis space admits no closed-form or finite-ensemble surrogate—a posterior defined directly over natural-language hypotheses or program text rather than a hand-specified low-dimensional model—thereby excluding tractable-model proxies such as GNoME's deep-ensemble policy.

Architecturally, with θ implicit in frozen weights and no per-experiment update, a system must either maintain an explicit probabilistic surrogate (GNoME's route, viable only where a calibrated input–outcome model exists) or use amortized EIG estimation [Foster et al., 2021], which remains unsolved for open-ended $\mathcal{S}$. Part of this gap is moreover conceptual: EIG-based selection presupposes an already-represented hypothesis space, whereas how genuinely new hypotheses or representations *arise* falls outside current BED formulations [Yanai and Lercher, 2019]—a generative limitation complementary to the channel-based non-identification discussed in Appendix A.

7.3 The Experiment Fidelity Gap

Test-time compute has created a structural asymmetry: hypothesis generation is now fast and cheap, while experiment execution E remains physically rate-limited—more reasoning cannot reduce DFT wall-time, wet-lab turnaround, or synthesis lags. Yet no surveyed system has a principled criterion for when to escalate from computational proxy E to physical experiment; result-caching accumulates past outcomes but does not determine *when* to invest in higher-fidelity measurement. Without an escalation criterion, computational loop closure does not imply physical discovery.

The pattern is consistent across domains. In materials science, GNoME predicts 380,000 stable compounds via DFT [Merchant et al., 2023], but only 736 match structures independently present in or added to the ICSD, of which 184 are new since the project began [Merchant et al., 2023]—these record concurrent work by others rather than syntheses the system triggered: the escalation decision is made externally with no principled automation. In chemistry, ChemCrow and ChemReasoner close the computational loop but leave the wet-lab loop open: NMR and MS characterization require physical execution that no current system knows when to invoke [Gromski et al., 2019]. In biology, BioPlanner's V metrics are insufficient to close the loop autonomously [O'Donoghue et al., 2023]—the physical execution question is never reached. In cognitive science, Jagadish et al. [Jagadish et al., 2026] use a behavioral foundation model as proxy E (synthetic participant data in lieu of human subjects) but give no criterion for escalating to in-vivo validation, themselves flagging synthetic data as a hypothesis accelerator, not ground truth. GNoME is the most successful case precisely because it *fixes* E at a single fidelity (DFT) and optimizes within that constraint: it circumvents the escalation problem rather than solving it. ERA likewise optimizes within a fixed computational evaluator, generating expert-level empirical software judged by a held-out quality metric without physical escalation [Aygün et al., 2026].

This problem is structurally distinct from Op1 and Op2: any V can evaluate only evidence at the fidelity it receives, and an EIG-based π can select only among the experiments represented at the currently available fidelity—so without E escalation criteria the system can close a computational loop indefinitely without reaching the physical evidence trustworthy scientific claims require.

Recent autonomous "self-driving" laboratories bear directly on this gap but do not close it. In materials science, the A-Lab [Szymanski et al., 2023] couples computational screening to robotic synthesis, initially reporting 41 of 58 target compounds synthesized, later revised after post-publication diffraction re-analysis to 36 of 40 reported successes confirmed (4 inconclusive) [Szymanski et al., 2026]—but it escalates by a fixed screen-then-synthesize rule rather than an evidence-informed criterion for when physical validation is warranted. Robotic chemistry platforms such as ORGANA [Darvish et al., 2024] automate the physical execution of wet-lab experiments, and self-correcting multi-agent systems such as AutoLabs [Panapitiya et al., 2025] automate the generation of hardware-ready protocols; reviews document the rapid maturation of the underlying hardware [Tobias and Wahab, 2025]. Yet automating E supplies capacity, not a criterion: these systems operate at a fixed or externally chosen fidelity and do not decide *when* accumulated computational evidence warrants escalating to a physical assay. The bottleneck is thus orthogonal to hardware automation—it persists even as the physical loop becomes mechanically executable.

Proposed falsification test: The gap is refuted if a system implements an evidence-informed criterion for escalating E from computational proxy to physical experiment and achieves a higher rate of experimentally validated discoveries per unit of physical experimental cost than a fixed-escalation system, evaluated on a pre-registered benchmark.

8 Discussion

Validator design as a diagnostic bottleneck. Our analysis suggests that unreliable validation can prevent a system from learning whether its outputs represent progress—an ecosystem-scale symptom is the documented surge in fabricated citations [Zhao et al., 2026], and strong task performance need not imply an internalized world model [Vafa et al., 2025]. Since the surveyed systems do not isolate V from G, E, M, and π, we read V completeness as a recurring diagnostic bottleneck, not a universal necessary condition or causal priority. This hypothesis-generating framing suggests a question rather than an evidence-backed recommendation: what level of V completeness does a domain support, and how is that signal used by G and π?

Human oversight. Rather than treating $H \in [0, 1]$ as a quantity to minimize, we read explicit human integration at the V_{human} and π levels as appropriate for the surveyed systems: their automated proxies do not yet provide independently validated novelty and significance judgments. This is a current-workflow finding, not a claim that those judgments are non-automatizable in principle. The goal is not full autonomy but *appropriate* autonomy.

Towards collective AI co-scientists. All systems analyzed are single-agent loops; extending Servo to a multi-agent collective with shared memory w and a collective validator [Taniguchi et al., 2025] poses exactly the three open problems above as its core implementation challenges, with ARA [Liu et al., 2026] a plausible entry point for w. Such a collective must also preserve epistemic diversity: AI reviewers already overlap far more than humans [Kim et al., 2026], risking the transient diversity that benefits epistemic communities [Zollman, 2010].

Limitations. Servo abstracts away several factors: the cost structure of E (orders of magnitude across domains), the non-stationarity of S, and the single-agent framing's social layer [Taniguchi et al., 2025]. More fundamentally, the V-completeness pattern is a cross-system *association*, not an established causal determinant, and the surveyed systems are a convenience sample subject to selection and survivorship bias. Mechanical loop closure moreover does not require completeness (AI Scientist v1 and NovelSeek close a computational loop at low completeness), so the claim concerns *trustworthy* closure, which we do not measure independently of V. The coding methodology and multi-vendor reliability estimates are in Appendix B (calibration Fleiss' κ=0.79; holistic V-completeness ordinal only κ=0.39; no calibrated *automated* novelty gate). Using LLM coders to characterize systems whose validators we argue are unreliable is a reflexive limitation, which is why the central claim is pinned to the calibration sub-construct and human expert coding remains necessary future work.

9 Conclusion

We introduced Servo (S, G, E, V, M, π), connecting AI Scientist systems to POMDP and BED theory across six core systems and seven domains. Its analytical distinctions support only narrow diagnostic claims, and the organizing hypothesis linking gating calibration to trustworthy closure remains provisional. Servo names the vocabulary for closing the remaining bottlenecks: validator completeness, search-policy design, and experiment-fidelity escalation.

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Conference For AI Scientists 2026 - AI Involvement Checklist

This checklist documents the role of AI systems at each stage of this research. The checklist has two parts: a **Research Stage Assessment** (items 1–4) rating AI involvement and iteration effort, and an **AI System Documentation** section (items 5–6).

Research Stage Assessment

1. **Hypothesis development.**
   Answer: **[B]**
   Iteration Effort: **[Medium]**
   Explanation: The core Servo framework tuple (S, G, E, V, M, π), the V-completeness thesis, the H parameter, and the three open problems were conceived by the human author. A frontier LLM agent elaborated framework components, identified literature-level instantiations, and proposed theoretical connections. The human author directed all conceptual decisions and made several substantive corrections to the AI's elaborations.
2. **Experimental design and implementation.**
   Answer: **[B]**
   Iteration Effort: **[Medium]**
   Explanation: This submission contains no computational experiments. The methodology for cross-system analysis (system selection criteria, component extraction protocol, domain comparison structure) was designed by the human author. A frontier LLM agent executed literature retrieval, extracted framework components from primary sources, and performed citation-level fact verification across four verification sweeps.
3. **Analysis of data and interpretation of results.**
   Answer: **[C]**
   Iteration Effort: **[Medium]**
   Explanation: The LLM agent extracted Servo components for all analyzed systems from primary sources and performed citation-level fact verification (4 verification sweeps). The human author reviewed all extractions, corrected mischaracterizations, and synthesized the cross-domain pattern. The V-completeness thesis emerged from the human author's synthesis. Consistent with this venue's premise, the cross-system coding is AI-conducted under human direction and disclosed as such; its rigor rests on auditability (released rubric, per-row source quotes, and regeneration/citation-validation scripts) and conservative claim-scoping rather than an independent human gold standard, which we treat as orthogonal to the framework contribution and defer to future work.
4. **Writing.**
   Answer: **[C]**
   Iteration Effort: **[High]**
   Explanation: The LLM agent drafted the majority of the manuscript text through multiple revision cycles; the human author provided key conceptual passages (framework definitions in §4, core open problems in §7), directed structural decisions, reviewed all drafts, and made final editorial judgments.

AI System Documentation

5. **AI systems used.**
   Description: A frontier large language model with code execution and long-document processing capabilities was used as the primary research and writing assistant. The system operated as an interactive CLI agent with file-system access, able to read primary sources, execute and debug Python simulations, and iteratively revise LaTeX manuscripts.
6. **Observed AI Limitations.**
   Description: (1) Systematic citation fabrication: the agent generated plausible but non-existent quotes and statistics, requiring four dedicated citation-verification sweeps using parallel sub-agents. (2) Analytical overconfidence: the agent initially overclaimed empirical significance for mechanically-entailed theoretical consequences; multiple adversarial

review rounds were needed to correct framing and remove overclaims. (3) Context drift: over long sessions, the agent occasionally regressed previously corrected errors, requiring explicit re-verification of integrity items before each commit.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction explicitly state the three contributions (framework, cross-domain analysis, open problems). The bounded analytical observations in §A clarify the separation of observation, reward validation, and novelty judgment rather than establishing a V-completeness theorem, and the cross-domain survey is presented as illustration; the paper delimits both the statistical conditions and the survey's illustrative role in §A and §8.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Limitations are discussed in §8 (framework abstracts domain cost structures, social layer, non-stationarity) and throughout the domain analysis (§6).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

   Answer: [NA]

   Justification: The paper does not claim a new theorem or proof. Appendix A states three bounded analytical observations, identifies the extra conditions needed to invoke standard results on posterior consistency and model misspecification, and expressly leaves adaptive open-ended regimes outside their scope. The cross-domain survey illustrates the distinctions rather than serving as a proof.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: This paper contains no computational experiments. The cross-domain analysis approach (system selection and component extraction from primary sources) is described in §5 and §6; machine-readable extraction sheets (core systems and the seven-domain Table 2) with a representative source quote and citation per row and citation-validating regeneration scripts, explicit inclusion/exclusion criteria, multi-vendor blind coding over 14 systems with Fleiss' κ (plus a two-vendor Cohen's κ and recode sensitivity for the calibration re-coding), and a counterexample search are released as supplementary material (see §8). All analyzed papers are publicly accessible. The released coding should be treated as *provisional structured annotation*: the table-generation scripts verify citation presence and non-empty source quotes but do not validate semantic entailment of each coded field; independent reproduction of the interpretive coding requires consulting the cited primary sources directly.

5. **Open access to data and code**

   Question: Does the paper provide open access to data and code?

   Answer: [Yes]

   Justification: The reproducibility package is released as supplementary material in the `analysis/` directory: the coding protocol (`coding_protocol.md`); coding sheets with a representative source quote and citation per row for both the core systems (`systems.csv`) and the seven-domain Table 2 (`domain_systems.csv`); the multi-vendor calibration re-coding (`multicoder/`); the reliability report (`reliability_report.md`); and the citation-validating table-regeneration scripts (`build_servo_tables.py`, `build_domain_tables.py`, `compute_calib.py`). All analyzed papers are publicly accessible; no separate dataset was generated. The complete reproducibility package is additionally released at `https://github.com/ysys143/servo-caisc2026`.

6. **Experimental setting/details**

   Question: Does the paper specify all details necessary to understand the results?

   Answer: [Yes]

   Justification: The cross-domain analysis methodology is described in §5 and §6. No computational experiments were performed in this submission.

7. **Experiment statistical significance**

   Question: Does the paper report error bars or statistical significance information?

   Answer: [NA]

   Justification: This paper contains no computational experiments and reports no statistical results requiring error bars or significance tests.

8. **Experiments compute resources**

   Question: Does the paper provide sufficient information on compute resources?

   Answer: [NA]

   Justification: This submission contains no computational experiments. The meta-analysis requires only standard document processing and no specialized compute.

9. **Research Ethics**

   Question: Does the research conform with the NeurIPS Code of Ethics, except where CAISc's AI authorship policies apply?

   Answer: [Yes]

   Justification: This is a meta-analysis of published systems with no human subjects, no proprietary data, and no safety risks. AI authorship is documented per CAISc policy.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive and negative societal impacts?

    Answer: [Yes]

    Justification: §8 addresses both the potential for accelerating scientific discovery and the risk of premature convergence under widespread AI Scientist deployment (particularly via the experiment fidelity gap and the current lack of validated automated novelty gates).

A Analytical Separation of Observation, Reward, and Novelty Channels

We separate three objects that are easily conflated: the *observation likelihood* $p(o \mid \theta, d)$ used for Bayesian updating, the *validator* $V(o)$ used as a reward or acceptance signal, and any additional channel that carries evidence about novelty or significance. The observations below apply standard results to fixed idealized settings. They are not new general theorems for adaptive experimental design or open-ended science.

Observation 1: posterior consistency belongs to the observation channel. In a dominated, fixed i.i.d. model $\{p_\theta : \theta \in \Theta\}$ with true distribution $p_{\theta^\star}$, posterior consistency can follow when the model identifies the truth, the prior puts positive mass on every Kullback–Leibler neighborhood of it, and suitable consistent tests separate it from complements of its neighborhoods, together with the regularity conditions needed by the relevant consistency theorem [Ghosal and van der Vaart, 2017]. Compactness, identification, and prior support stated only in a topological sense are not substitutes for those conditions. The cited result therefore does not establish consistency for the adaptive, non-stationary design sequence of an AI Scientist without an additional theorem for that regime.

Equation 1 defines a one-step information objective. Maximizing it is myopically optimal for that objective by construction, but this does not establish global optimality for a sequential scientific policy. Let $\mathcal{H}_{t-1}$ contain the past designs and observations, let d_t be measurable with respect to that history, and define $\text{EIG}_t = I(\theta; O_t \mid \mathcal{H}_{t-1}, d_t)$. Posterior concentration does not require a divergent sum of these one-step values. For example, when θ is discrete with finite entropy, the conditional-information chain rule gives

$$\sum_{t\geq 1} \mathbb{E}[\mathrm{EIG}_t] = I(\theta; \mathcal{H}_\infty) \leq H(\theta), \tag{4}$$

so finite cumulative expected information is compatible with posterior concentration. If V remains outside the likelihood, it has no direct role in this fixed-model consistency statement, although policy use of V can indirectly change which future designs and observations occur.

Observation 2: a coupled validator changes the pseudo-true target only through a normalized likelihood. For a fixed design d, suppose an update uses

$$q_{\theta,d}(o) = \frac{p(o \mid \theta, d)\exp\{\beta V(o)\}}{Z_\theta(d)}, \qquad Z_\theta(d) = \int p(u \mid \theta, d)\exp\{\beta V(u)\}\, du < \infty. \tag{5}$$

Let F denote the fixed i.i.d. data-generating distribution. Under the regularity, integrability, separation, and prior-mass conditions required for a Berk-type misspecification result, posterior mass is driven toward a minimizer set rather than automatically toward a unique point:

$$\Theta^\dagger = \arg\min_{\theta\in\Theta} \mathrm{KL}(F \,\|\, q_{\theta,d}) \tag{6}$$

Only when this set is a singleton do we denote its member by $\theta^\dagger$ [Berk, 1966]. Validator-induced drift means that this minimizer set differs from the corresponding projection under $p(\cdot \mid \theta, d)$; an outcome-dependent V can alter the normalizer $Z_\theta(d)$ even when V has no explicit θ argument, but non-constancy alone neither guarantees nor is required to prove a target change. An additive constant in V cancels between numerator and normalizer, while $\beta = 0$ gives $q_{\theta,d} = p(\cdot \mid \theta, d)$. A validator used only by the policy may still alter future design selection, but that is an adaptive-design effect, not the fixed-i.i.d. conclusion of Berk [1966].

Observation 3: novelty invisible to every available channel remains unidentified. Write the latent state as $\theta = (\theta_{\mathrm{nov}}, \psi)$ and suppose the prior factorizes as $\Pi(d\theta_{\mathrm{nov}}, d\psi) = \Pi_{\mathrm{nov}}(d\theta_{\mathrm{nov}})\Pi_\psi(d\psi)$. If the joint likelihood of all observed channels depends on ψ but not on θ_{nov}, then the posterior marginal of θ_{nov} equals Π_{nov} for every dataset. This full factorization from every likelihood-relevant latent component is essential: independence only from already identified components would not exclude indirect updating through a prior-correlated nuisance variable. The novelty component is therefore not consistently identifiable from those channels. Identification requires an additional informative channel, but the premise does not imply that the channel must be human in principle. The survey instead supports the narrower empirical statement that novelty and significance judgments in the systems examined here remain human-mediated or lack a validated automated gate.

These observations support the paper's diagnostic separation of observation, reward validation, and novelty judgment. They do not prove that the empirical V-completeness ordinal is necessary for all trustworthy autonomous discovery, establish causal effects of validator design, or settle the paper's open-ended and adaptive setting. Those broader claims remain provisional framework hypotheses to be evaluated with independently coded outcomes and domain-specific evidence.

B Cross-System Coding Reliability

The cross-system survey is released as provisional structured annotation. The claim concerns *trustworthy* closure, which we do not measure independently of V: the `trustworthy_closure` coding draws on outcome signals only partly outside V (peer review and publication overlap V_{human}) and, at six systems with outcomes entangled with V, cannot test the direction. For traceability we release the rubric, coding sheets with a representative source quote and citation per row (core systems and the cross-domain Table 2), and validator scripts that reject any uncited or unquoted row; the scripts verify formal presence of quotes and citation keys but not semantic entailment, so per-field judgment still requires the cited primary source.

Three independent, model-pinned, blind LLM-based coder agents from different vendors coded 14 Tier-1 systems (see `multicoder/target_systems.md`): Claude Code (`claude-opus-4-8`, Anthropic), Codex CLI (`gpt-5.5`, OpenAI), and Antigravity (`gemini-3.1-pro`, Google), each run headless and model-pinned, with exact invocations and per-model outputs released in the supplement. The six core systems in Table 1 are a representative analytical sample spanning the discovery-loop spectrum. Inter-coder agreement (Fleiss' κ over the 13 systems coded by all three vendors—NovelSeek drew only two usable codings after a coder timeout and is excluded) is substantial-to-perfect on most constructs (calibration $\kappa{=}0.79$, loop status $\kappa{=}0.74$, semantic-validation presence $\kappa{=}1.0$; mean $\kappa{=}0.68$) and only fair on the holistic V-completeness ordinal ($\kappa{=}0.39$)—the one construct the central claim does not rest on (it rests on the calibration sub-construct, $\kappa{=}0.79$). Agreement between the coders and the author's reference labels was originally negative on that ordinal (-0.25 to -0.43); a later audit found part of that gap to be author coding error rather than construct noise, and after correction it is 0.17 to 0.67 (Appendix D). The coder-only figures quoted here are unaffected by that correction. All reported as automated transparency, not human-level reliability. The low ordinal agreement plausibly reflects that a single ordinal compresses distinct properties—gate calibration, property coverage, experimental fidelity, and external independence—that a vector-valued treatment would separate; we therefore read the V-completeness ordinal as exploratory rather than a central measure.

A counterexample search found no V-incomplete system achieving *trustworthy* closed-loop discovery (the lone near-miss, AI Scientist v1, closes on a biased validator but yields unreliable output). A two-vendor recode over all 14 systems (supplement) resolves a *present*-vs-*gating* ambiguity in the calibration construct—the reference labels and both coders agree the layer is present but non-gating (raw agreement 13/14, Cohen's $\kappa{=}0.86$; a robustness check, not expert-validated reliability)—and finds no system with a calibrated *automated* novelty gate, grounding the novelty open problem. Introduced post-hoc, this $V_{\text{present}}/V_{\text{gating}}$ split reflects iterative construct development rather than a prospective reliability study.

C Per-Domain Analysis

This appendix expands Table 2 into a per-domain reading in the Servo vocabulary, stating for each field the validator type and its reliability, the loop status that validator permits, the search policy actually deployed, and the bottleneck that follows. The coding is released as the machine-readable `domain_systems.csv`, with a representative source quote and citation keys per row; the entries below paraphrase those quotes and should be read as a structured reading of the cited work rather than an independent empirical study.

Mathematics, formal (V_{formal}; fully closed). FunSearch [Romera-Paredes et al., 2024] and DeepSeek-Prover [Xin et al., 2024] validate against a formal or numeric oracle—an evaluator function or proof kernel returning a deterministic pass/fail. This V_{formal} provides deterministic formal-correctness feedback, enabling mechanically closed evaluation within the formalized task; the loop is fully closed. The residual bottleneck is not formal correctness but S: pretraining contamination makes it hard to certify that a discovered result is genuinely novel rather than memorized—the formal-domain instance of the channel-based novelty problem discussed in Appendix A.

Mathematics, natural language ($V_{\text{sem}} + V_h$; closed at low quality). Aletheia [Feng et al., 2026] produces natural-language proofs scored by LLM and human judgment. The loop closes mechanically, but V_{sem} is imperfect: of 200 audited Erdős solutions only 13 (6.5%) were rated meaningfully correct. Replacing the formal oracle with a semantic one reintroduces human review and exposes hallucination and plagiarism as failure modes. This is evidence about the reliability of the reward or acceptance channel; without a specified normalized update likelihood it is not an instance of the fixed-i.i.d. coupling analysis in Appendix A. The Leiden Declaration [Alper et al., 2026] corroborates the practical limitation: AI-produced natural-language proofs may contain almost-invisible errors that resist detection without expert scrutiny. This supports continued expert review in current workflows, not a theorem that human validation is irreplaceable in every possible future system.

Physics ($V_{\text{emp}} + V_h$; none/partial). AI Feynman [Udrescu and Tegmark, 2020] fits equations to data with deterministic search, while symbolic-model discovery [Cranmer et al., 2020] distills equations from trained neural networks via stochastic symbolic regression (genetic algorithms). Validation is empirical and of medium reliability, but the systems carry no memory transfer across problems: each task starts fresh. The bottleneck is thus the absence of cross-problem memory transfer: each task begins from scratch without accumulated experimental patterns from prior discoveries.

Chemistry (V_{emp} surrogate; computational loop only). ChemReasoner [Sprueill et al., 2024] validates computationally with surrogate signals and searches by beam, whereas ChemCrow [Bran et al., 2024] plans via ReAct tool-use and validates in part through physical wet-lab execution on a robotic platform (with expert and LLM evaluation). The computational loop closes, but physical characterization (NMR, MS; [Kamber, 2026]) leaves the wet-lab loop open: the validator that would confirm a real discovery sits outside the automated boundary, so closure is partial at the measurement interface.

Materials science (V_{emp} via DFT; computationally closed). GNoME [Merchant et al., 2023] runs DFT-in-the-loop active learning, predicting 2.2 million structures stable with respect to prior work (380,000 on the updated convex hull), with uncertainty estimates that the manuscript reads as a coarse approximation to the BED ideal within the computational surrogate. DFT should be read as an approximation of physical energies rather than a fully calibrated empirical oracle for materials discovery: of the predicted stable structures, 736 match entries independently present in or added to the ICSD, of which 184 are new since the project began [Merchant et al., 2023], and critics note that predicted compounds should not be equated with validated materials exhibiting demonstrated functionality [Cheetham and Seshadri, 2024]. Servo therefore codes this domain as computationally closed for phase-stability screening, with the physical materials-discovery loop only partly closed. The remaining bottlenecks are resource-bound: DFT wall-time and the lag before predicted structures are synthesized and characterized.

Biology ($V_{\text{emp}} + V_h$; partial). BioPlanner [O'Donoghue et al., 2023] generates experimental protocols, but the validation signals are weak and biological-validity metrics are insufficient to close the loop autonomously, so reliability is low-to-medium with effectively no search policy beyond generation. The bottleneck is V itself: without metrics that track ground-truth biological validity, the loop cannot be closed on anything trustworthy.

Social science (statistical tests; partial). Automated-social-science pipelines [Manning et al., 2024] validate with statistical tests over full-factorial designs, while LLM-simulation studies [Aher et al., 2023, Park et al., 2023a] validate simulated behavior with statistical tests over ablation and controlled comparisons. The characteristic failure is circular validity: when the generator G and the simulated subjects share a model, the validator measures the model's self-consistency rather than an external fact—an identifying-channel problem of the kind separated analytically in Appendix A.

Computer science / algorithms (V_{emp}; fully closed). AutoML-Zero [Real et al., 2020], algorithm discovery [Surina et al., 2025], and AutoKernel [Jaber and Jaber, 2026] close the loop via reliable benchmark/empirical validation and evolutionary or LLM-guided search. Validation is high-reliability and the loop is fully closed; the residual bottleneck is interpretability—the discovered artifacts work but resist human understanding, a limitation of M and output form rather than of V.

Cross-domain reading. In the illustrative coding, observed bottlenecks vary with V type alongside the other framework components: formal oracles support mechanically closed evaluation within formalized tasks; imperfect semantic or empirical validators co-occur with hallucination, circular-validity, or insufficient-metric failures; and expensive empirical evaluation in materials science co-occurs with a resource bottleneck. This is the diagnostic use of Table 2: it does not claim a tested cross-domain law or isolate a causal effect of V, but shows that the same (S, G, E, V, M, π) vocabulary localizes where each domain's loop breaks. V type is one factor associated with trustworthy closure, while the search policy π separately bears on which represented regions of S the loop explores. Table 2 captures the V dimension of this provisional two-axis reading; the π dimension remains underdeveloped across the surveyed domains, with materials science a partial exception.

D Post-Submission Revisions

This manuscript continues to be refined after the CAISc 2026 submission deadline. This appendix is the running record of every difference between it and the archival version of record, so that the two can be reconciled entry by entry. Each entry states what changed, why, and whether the version of record is thereby wrong or merely less precise.

R1. Leiden Declaration signatory count (Section 7). *Status in the version of record: unsupported by the source cited for it.* The version of record states that the Leiden Declaration was "signed by over 130 mathematicians," attributing that figure to the declaration document. The document states no signatory count at all: its only numerals are section numbers and the year of the meeting that produced it. The count is maintained on the separate signature site, which recorded 3,071 signatories when checked on 20 July 2026. The defect is therefore not that the figure was understated but that it was attributed to a text that does not contain it. This revision cites the signature site for the count, marks it as of a date because it grows over time, and notes that the declaration itself gives none. The substantive claim the sentence supports—that AI-produced proofs may carry errors that resist detection without expert scrutiny—is stated in the declaration and is unaffected.

R2. Formal validators and RL stability (Section 6). *Status in the version of record: overclaim; the sentence asserts more than the evidence supports.* The version of record states that formal mathematics has "a binary V_{formal} that makes RL stable and the loop fully closed." A deterministic pass/fail oracle settles whether a candidate proof or program is acceptable, but the stability of policy learning also depends on reward density, the state space, the policy update rule, the exploration strategy, and function approximation; the surveyed formal systems do not isolate the validator as the cause of stable training (DeepSeek-Prover-V1.5 additionally uses supervised fine-tuning and an MCTS variant with intrinsic rewards, and FunSearch couples the evaluator to an LLM and an evolutionary procedure). The clause is replaced here with "enables mechanically closed evaluation within the formalized task," matching the wording already used for the same systems in Appendix C. No other claim depends on the removed clause.

R3. Scoring of novelty, significance and correctness (Section 7). *Status in the version of record: contradicted by the cited source.* The version of record cited "The AI Scientist's 1–10 score" as an instance of collapsing the three properties into one scalar. The cited system in fact rates ideas on *separate* interestingness, feasibility and novelty scales, and its reviewer emits several axes; the collapse occurs not in scoring but at the accept/reject decision, which thresholds a single aggregate. The sentence now locates the collapse where it happens. The open problem is unchanged.

R4. "Greedy" as a coding, not a source claim (Section 7). *Status in the version of record: unsupported interpretation.* The cited source describes archive-conditioned idea generation but defines no argmax, best-score or greedy selection rule; "greedy" is this paper's own coding. The sentence now says so.

R5. Two different quantities compared (Section 5). *Status in the version of record: contradicted in metric meaning.* The version of record read "balanced accuracy 0.65 vs. human 0.66," which invites reading 0.65 as agreement with human reviewers. It is the balanced accuracy of predicting ICLR 2022 accept/reject labels; 0.66 is human inter-reviewer consistency measured on a different pool. The two are now distinguished.

R6. Class definition and Agent Laboratory (Introduction). *Status in the version of record: contradicted by the cited source.* The opening sentence defines the class as agents that *independently* generate hypotheses and cited Agent Laboratory among them; in that system the research idea is supplied by a human. The citation has been removed from the class-defining list. Agent Laboratory remains analysed throughout the paper.

R7. Robot Scientist table cells (Table 1). *Status in the version of record: one unsupported, one mislabelled.* The novelty-measure cell read "Stat. test"; the statistical test in the cited source adjudicates growth phenotypes and gene-function hypotheses, not scientific novelty, so the cell now reads "None (corr.)." The policy cell "Active learn." is not supported by the 2010 review used for that column; the caption now attributes the active-learning policy to the primary source that documents it.

R8. What "calibration" denotes (Section 4). *Status in the version of record: a term used in a weaker sense than the word normally carries, without saying so.* The version of record calls a validator "uncalibrated" on the strength of discrimination statistics. Calibration and discrimination are independent properties: balanced accuracy, F1 and AUC measure how well a scorer separates classes, while calibration—whether stated confidence matches empirical frequency—is measured by reliability curves, expected calibration error or Brier score. No surveyed source reports the latter, so neither "calibrated" nor "uncalibrated" is established in the strong sense. The paper's own coding protocol already defines the field operationally as *not systematically biased* rather than as probabilistically calibrated. This revision states that definition in the body where the term is first made operative. The construct, its coding, and the reported inter-coder agreement are unchanged; only the term's scope is now explicit.

R9. The automated reviewer is not a gate (Section 5). *Status in the version of record: contradicted by the cited source.* The version of record described the automated reviewer as the system's "internal acceptance gate." In the cited source the reviewer scores finished manuscripts and configurations; what advances the search is a stage evaluator, a best-first policy and a compute budget, and the manuscripts sent for external review were chosen by hand. The reviewer is an evaluation instrument, not a gate. The sentence now says so, and the acceptance bias it does exhibit is stated with the figure that supports it (false-positive rate 0.45 against 0.17 for human reviewers).

R10. Coding corrections and their effect on reported agreement (Appendix B). *Status in the version of record: reference labels contained demonstrable errors, and one reported statistic depended on them.* An audit of the released coding sheet against the paper's own protocol found eleven cells that the protocol, the primary sources, or the manuscript itself contradict. `Vsyntax` was coded 0 for every system, including five that execute code, against a rubric whose example of the field is "code compiles/executes"; the blind coders assigned 1 unanimously on four of them. `Vhuman` was coded 1 for AI Scientist (Nature 2026) on the strength of an external post-hoc workshop review, which the protocol's "required for the validity/novelty signal" test does not cover, and 0 for Agent Laboratory, contradicting this paper's own table, which records that system's novelty measure as human-delegated. `Vcompleteness` was 3 for AI Scientist (Nature 2026), a value the scale licenses only when the top automated layer is calibrated; that flag had been corrected to 0 earlier without propagating to the ordinal. The remaining ordinals were re-derived from the scale anchor after the `Vsyntax` correction rather than copied from the coders; that derivation independently reproduces the coder consensus in every case. *Effect on reported statistics:* the Fleiss agreements among the three blind coders, which are the figures this paper quotes, are identical before and after the correction, as are the pairwise coder agreements. What moves is agreement between the coders and the author's reference labels—necessarily, because those labels were revised. The released reliability report now states both the original and the corrected values and warns that the rise is an artefact of the revision rather than evidence of reliability.

R11. NovelSeek / InternAgent citation identity (Table 1). *Status in the version of record: bibliographic entry does not match the work as it now stands.* The entry keyed `zhang2025novelseek` gave the title and individual author list under which arXiv:2505.16938 first circulated. Later versions of the same preprint (v3, the one audited here) retitle the work *InternAgent* and place it under a laboratory team byline. The citation is to the right work—the identifier is unchanged—but a reader retrieving 2505.16938 today finds a different title and author line than the bibliography gives. The entry now carries the current title and byline, records the former name, and keeps the key so that earlier citations of this work still resolve. The manuscript continues to call the system NovelSeek, which is how it is named in the analysis released with this paper, and notes the retitling in the caption of Table 1, where the system is introduced.

R12. What the 736 GNoME matches are (Section 7). *Status in the version of record: the phrase is the source's own, but it invites a reading the source does not support.* The version of record wrote that of 380,000 predicted stable compounds "only 736 have been independently experimentally realized," which is verbatim from the cited abstract. The cited Methods make the mechanism explicit: the 736 are matches between GNoME's predicted structures and entries already present in, or added to, the ICSD when it was queried in January 2023, and 184 of them are new since the project began. They are therefore structures other groups made independently or concurrently, not syntheses the system prompted. Used as the numerator of a realization rate, the original phrasing suggests a predict-then-synthesize pipeline that did not operate. The sentence now states the mechanism. This strengthens rather than weakens the point it supports—that escalation to physical work is decided outside the system—since the matches were not triggered by any escalation criterion at all.

R13. Epistemic status of the formal claims (Abstract; Section 4; Appendix A). *Status in the version of record: less precise and materially overstates the scope of the mathematical* *contribution.* The version of record presents three numbered propositions, a bundled assumption, and proof sketches as if they established results for a tractable AI Scientist loop. The paper's purpose is a diagnostic framework, however, and the cited statistical results do not supply a new theorem for adaptive, non-stationary, open-ended search. This revision replaces that theorem-and-proof posture with three explicitly bounded analytical observations, removes the unused theorem environments and their cross-references, and states in the abstract, background, checklist, conclusion, and domain discussion that the appendix imports standard fixed-setting distinctions rather than proving the framework. The six-component framework, three-way closure distinction, and provisional survey claims are unchanged; their evidential status is made explicit rather than upgraded by mathematical presentation.

R14. Posterior consistency and cumulative EIG (Appendix A). *Status in the version of record: partly incorrect.* The version of record treats compactness, identification, prior support, and a bespoke design-separation clause as a standard sufficient condition for posterior consistency under Ghosal and van der Vaart [2017], without stating the Kullback–Leibler prior-mass and testing conditions used by the relevant classical results or proving an adaptive-design analogue. Its proof sketch also says that positive but summable one-step EIG leaves the posterior short of a point mass. That claim is false in general: for a discrete parameter with finite entropy, the conditional-information chain rule bounds total expected EIG by initial entropy even when the posterior concentrates. The revision states the fixed-i.i.d. conditions only schematically and with their scope, removes the unsupported adaptive conclusion, and describes Equation 1 as a one-step myopic criterion rather than a globally optimal scientific policy. This narrows the theoretical rhetoric without weakening the observation-versus-reward distinction, the empirical BED–practice gap, or its falsification test.

R15. Misspecified validator coupling and KL projection (Appendix A). *Status in the version of record: mathematically under-specified and broader than the cited result.* The version of record writes the coupled update only up to proportionality, invokes Berk [1966] without restricting the claim to its fixed-i.i.d. misspecification setting, assumes a unique KL projection, and gives "outcome dependence" as if it were a sufficient drift condition. The revision defines the finite normalizer $Z_\theta(d)$, fixes the design and data-generating distribution, states the needed regularity and prior-mass qualifications, and uses the KL-minimizer set $\Theta^\dagger$ unless uniqueness is separately available. Drift now means that this set differs from the projection under the original likelihood; an additive constant cancels and an outcome-dependent validator need not change the target. Policy-only effects are separated as adaptive design selection, not attributed to Berk's theorem. The correction preserves the paper's diagnostic warning about coupling while withdrawing any claim that the surveyed systems instantiate this asymptotic result.

R16. Novelty non-identification and the human channel (Section 7; Appendix A). *Status in the version of record: the non-identification premise is correct, but its human-only conclusion exceeds that premise.* If every observed likelihood is independent of an a priori independent novelty component, its posterior marginal remains the prior; those channels cannot identify it. What follows is the need for an additional informative channel, not a theorem that this channel must always be human or that human validation is irreplaceable in principle. The revision makes that logical boundary explicit and limits the empirical conclusion to the surveyed systems: their novelty and significance judgments are human-mediated or lack a validated automated gate. This leaves the novelty open problem and its falsification criterion unchanged while removing a universal impossibility claim.

R17. Table 1 reconciled with the coding sheet. *Status in the version of record: two rows disagreed with the released coding they summarise.* The novelty row read "None" for Coscientist, where the sheet codes a human gate, and "Human-deleg." for Agent Laboratory, where the sheet codes a biased automated gate—the latter also contradicting Section 5, which describes that system's reviewer as biased. The H row rendered Coscientist's 0.75 as "$\approx$1" while giving Agent Laboratory's 0.8 exactly, so one table rounded two neighbouring values by different rules. Both rows now reproduce `novelty_gate` and `A_H` verbatim and the row is renamed to name what it reports. This also settles the Robot Scientist cell left open by R7: the sheet codes a predefined novelty criterion, which is what the cell now reads.

R18. Residual claim boundaries and analytical premises (Sections 4 and 7; Appendix A). *Status in the version of record: partly incorrect, and the R13–R16 revision remained incomplete.* The version of record identifies a sequential policy with a one-step EIG maximizer, treats validator reliability and human review too categorically, and states the novelty posterior-invariance argument without excluding prior coupling through other likelihood-relevant latent variables. The first mathematical revision narrowed these claims but left several stronger English sentences in place and retained the insufficient independence premise. This revision makes the EIG rule explicitly myopic, factorizes the novelty prior from every likelihood-relevant latent component, limits human mediation to a finding about the surveyed workflows, and recasts cross-domain validator patterns as provisional associations rather than necessity or causality. The Korean source is narrowed to the same scope. These changes repair the logical propagation of R13–R16 without changing the six-component framework, the three open problems, or their proposed falsification tests.

R19. Correction status and gate semantics (Abstract; Table 1; Appendix B). *Status in the version of record: materially inaccurate in several claims, while the earlier post-submission citation and coding language remained insufficiently precise.* Appendix D records corrections to factual, coding, statistical, and mathematical statements, so readers should not be directed to the archival text alone when relying on a corrected claim. The notice now asks readers to pair the version-of-record citation with this dated correction and states that the correction should be linked from the official record when possible. Table 1 also called a biased automated score a "novelty gate" for AI Scientist (Nature 2026), although the paper and released coding set `V_gating=0`: the reviewer scores completed outputs but does not advance search or select external submissions. The row is therefore renamed "Novelty assessment," the non-gating status is explicit, and the caption distinguishes the legacy `novelty_gate` field from operational gating. Finally, the abstract withdraws the claim of direction consistency: because the six-system outcome coding partly overlaps V, even the direction of association remains untested. These corrections preserve the framework and hypotheses while removing unsupported evidential status.

Scope of revision. The entries above are the complete set of changes made since the archival deposit, R1–R19. Entries R3–R12 narrow or correct statements that the citation audit judged unsupported by, contradicted by, or less precise than their cited sources. Entries R13–R16 record the subsequent adversarial mathematical review: they replace the proposition/assumption/proof-sketch presentation with bounded analytical observations, correct the EIG statement, delimit the conditions under which the Ghosal–van der Vaart and Berk references may be invoked, and distinguish mathematical non-identification from the surveyed use of human judgment. R17 reconciles two table rows with the released coding sheet. R18 records the adversarial recheck that completed the propagation of the earlier mathematical corrections, including the stronger prior-factorization premise and the removal of residual universal, causal, and human-only wording. R19 corrects the citation guidance, distinguishes novelty assessment from operational gating, and withdraws the untestable direction-consistency statement. These changes do not alter the six-component framework, the three open problems or their falsification criteria, or the status of the cross-system survey as provisional structured annotation. The bibliography is unchanged apart from the single entry corrected in R11. Should this manuscript be revised further, later entries will be appended to the list above so that the full distance from the version of record remains reconstructible from this appendix alone.